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Top Skills

Tokenization
Program Monitoring
Event-driven

Languages

English (Full Professional)
Korean (Professional Working)
Hindi (Native or Bilingual)

Certifications

Claude Certified Associate -
Foundations
Architecting on AWS
AWS Certified Solutions Architect -
Associate
Big Data Fundamentals 101

Patents

Electronic apparatus and controlling
method thereof
Server Apparatus and Method for
providing search result thereof

Manmohan Singh Bohara

Software Development Engineer III at Tesco Technology specialising
in Cloud Computing
Cambridge, England, United Kingdom

Summary

At Tesco Technology as a Software Development Engineer III, my focus lies in pioneering cloud solutions that harness the power of Kubernetes and Azure DevOps. With a rich background in R&D at Samsung Electronics, innovation and technical leadership have been the cornerstones of my career. Together with my team, we've spearheaded projects that have significantly enhanced SmartTV ecosystem services, contributing to a more intuitive user experience.

My tenure in the industry has been marked by a dedication to collaborative growth and problem-solving. We have succeeded in deploying robust software solutions and embracing a culture of innovation that resonates throughout the company. My expertise in cloud computing is not just a professional skill set but a mission to drive forward the technologies that shape our digital world.

Experience

Tesco Technology

3 years 8 months

Software Development Engineer III

February 2023 - Present (3 years 8 months)

Welwyn Garden City, England, United Kingdom

Software Engineer

February 2023 - Present (3 years 8 months)

Developed and managed microservices-based APIs capable of handling high-throughput traffic (up to 5000 TPS) during peak loads. Designed a decoupled microservice architecture for promotion management, using event-driven communication (e.g., Kafka) to publish and consume promotional data, enabling horizontal scaling of high-traffic components independently from lower-traffic services. Built and deployed integrity services to validate invoice pricing accuracy by cross-verifying with cloud-based price management systems, creating metrics and alerts in monitoring tools (e.g., New Relic) for

discrepancy detection. Developed monitoring dashboards (e.g., New Relic, Splunk) to track SLOs for availability, latency, error rates, and error budgets, while monitoring the health of Kubernetes, database, and messaging clusters. Implemented contract testing to ensure consistent and reliable interactions between microservices using industry-standard tools (e.g., Pact). Automated deployment pipelines using CI/CD tools (e.g., Azure DevOps, Helm Charts) to streamline service deployment on Kubernetes clusters. Applied horizontal autoscaling strategies in Kubernetes to optimize resource usage and handle fluctuating workloads. Utilized cloud-native identity management (e.g., Azure Managed Identities) to reduce secret management overhead and enhance security.

Samsung Electronics

12 years 1 month

Staff Software Engineer

April 2015 - January 2023 (7 years 10 months)

Built a unified system for bringing in metadata from various content partners (CP) to a smart TV platform, which boosted partner participation by 12%. This system used a microservices architecture since each partner shared data in different ways—some uploaded via PUT APIs, others had REST APIs to pull, and each partner provides data in different format. Multiple microservices processed the incoming data, standardizing it into a unified schema and handling tasks e.g. CDN URL preprocessing for images and videos. This unified data then used by other microservices consumed by Samsung devices. Designed and developed a foundational architecture for services that cut down new service development time by 15%. It was a layered architecture; the front layer, called the OSB (Outer Service Bus), handles all incoming requests with responsibilities like authentication, authorization, throttling, and routing etc. Next is the service layer, where microservices and their data reside. Finally, there's the ISB (Inter Service Bus) layer, which syncs data between multiple microservices as needed. This setup streamlined development and made adding services much easier. Designed and developed server-less applications using AWS services that reduced the cloud infra cost by as much as 30% for some services. Designed and developed efficient and cost effective cloud native applications by incorporating AWS Well-Architected framework pillars. Worked as technical lead for three overseas teams in India, China and Poland.

Senior Engineer Research and Development

March 2015 - January 2023 (7 years 11 months)

South Korea

Senior Software Engineer

January 2011 - March 2015 (4 years 3 months)

Improved click rate on search results by 12% by implementing custom ranking algorithm for content search on Samsung's SmartTV. Improved back-end batches that are used to fetch CP's metadata, parse, and make it searchable and reduced the time taken to make new metadata searchable by a factor of approximately 15%. Designed and developed APIs to return a personalized search results using C# and .NET framework. Developed a custom tokenizer for Korean language to search using Korean consonant that reduced typing significantly for Korean users. Automation of monitoring and notification of infrastructure using python boto3 that helped teams to proactively resolve issues even before they become critical.

Engineer Research and Development

January 2011 - February 2015 (4 years 2 months)

Joined as a Software Engineer in Samsung Electronics Head Quarter, South Korea. Has been working in various projects of SmartTV ecosystem including content search, recommendation and conversation services.

Nokia Siemens Networks

Senior Software Engineer

July 2007 - November 2010 (3 years 5 months)

Bangalore, India

Worked as a Senior Software Engineer in Nokia Siemens Networks (NSN), Bangalore, India.

In NSN, I mainly worked in various projects in Transport, Mobile and Fixed Network Management Applications (NMS), in functional areas like Software Management, Performance Management, Configuration Management and Topology Management.

LG Soft India

Associate Engineer

May 2006 - June 2007 (1 year 2 months)

Bangalore, India.

Worked as Associate Software Engineer in LG Soft India.

LG Electronics

Software Engineer

January 2006 - March 2007 (1 year 3 months)

Worked on IrisAccelerator project and helped team to improve the performance of IrisAcceleator by refactoring code.

Education

International Institute of Information Technology Bangalore
Executive Post Graduate Programme, Software Development · (January 2022 - March 2023)

Udacity

Nano Degree, Deep Learning · (2019 - 2019)

Kumaon Engineering College

Bachelor of Engineer, Electronics and Communications
Engineering · (2001 - 2005)

International Institute of Information Technology Bangalore
Executive Post Graduate, Computational Science

Kumaon

Bachelor of Engineering, Electrical, Electronics and Communications
Engineering